VoltPark

- **Theme:** SDG 11 – Sustainable Cities and Communities (Also aligned with SDG 13 – Climate Action & SDG 9 – Industry, Innovation & Infrastructure)
- **Category-** Software
- **Group No -** 17
- **Group Members-** Neev Rambhiya, Raj Dabholkar, Naman Shah
- **Guide-** Prof. Kanchan Wankhede

Proposed Solution

- VoltPark is a smart web-based platform to find, check availability, and pre-book parking and EV charging slots in urban areas.
- It eliminates manual searching for parking and charging, thereby reducing time waste, traffic congestion, fuel loss, and user frustration.
- Users can check availability before reaching the location and reserve a slot in advance for a specific date and time.
- The system provides separate dashboards for users, parking/charging owners, and administrators for efficient management.
- The solution is innovative because it combines parking and EV charging in a single platform with a pre-booking facility.
- It uses digital booking verification (QR/Code) and is scalable for future smart city and IoT integration.

Technical Approach

- The system will be developed using the *MERN Stack*:
  - MongoDB – for storing user, location, booking, and payment data
  - Express.js – for building backend APIs and handling requests
  - React.js – for building the user interface of the web application
  - Node.js – for server-side logic and system operations
- The application follows a client–server architecture, where:
  - Frontend (React) handles user interaction
  - Backend (Node + Express) handles business logic and data processing
- Google Maps API will be used for location display and navigation.
- Online payment gateway (test mode) will be integrated for booking payments.

Feasibility and Viability

- The proposed system is technically feasible using the *MERN Stack* and existing web technologies.
- It can be implemented in a phased manner, starting with a small number of parking and EV charging locations.
- The main challenges include:
  - Collecting and maintaining accurate availability data
  - Onboarding parking and charging station owners
- These challenges can be handled by:
  - Starting with manual or semi-automatic updates
  - Using a simple admin and owner dashboard for management
- The project is economically viable as it can generate revenue through bookings, subscriptions, and promotions.
- The system is scalable and can be extended in the future with IoT sensors and smart city integrations.

Impact and Benefits

- **Impact :**
  - Reduces traffic congestion and time wasted in searching for parking and EV charging.
  - Supports EV adoption by making charging infrastructure easy to find and access.
  - Contributes to smarter and more organized urban mobility.

- **Benefits :**
  - Saves time and fuel for users, improving daily travel experience.
  - Improves utilization and revenue for parking and charging station owners.
  - Helps in reducing pollution and emissions, supporting sustainable cities.

Research and References

- **Research Papers :**
  - https://arxiv.org/abs/2009.13443
  - https://www.mdpi.com/1424-8220/16/6/931
  - https://ieeexplore.ieee.org/abstract/document/8125048/

- **Reference :**
  - https://parkable.com/parking-management-system
  - https://www.mantratec.com/Solutions/Parking-Management-System